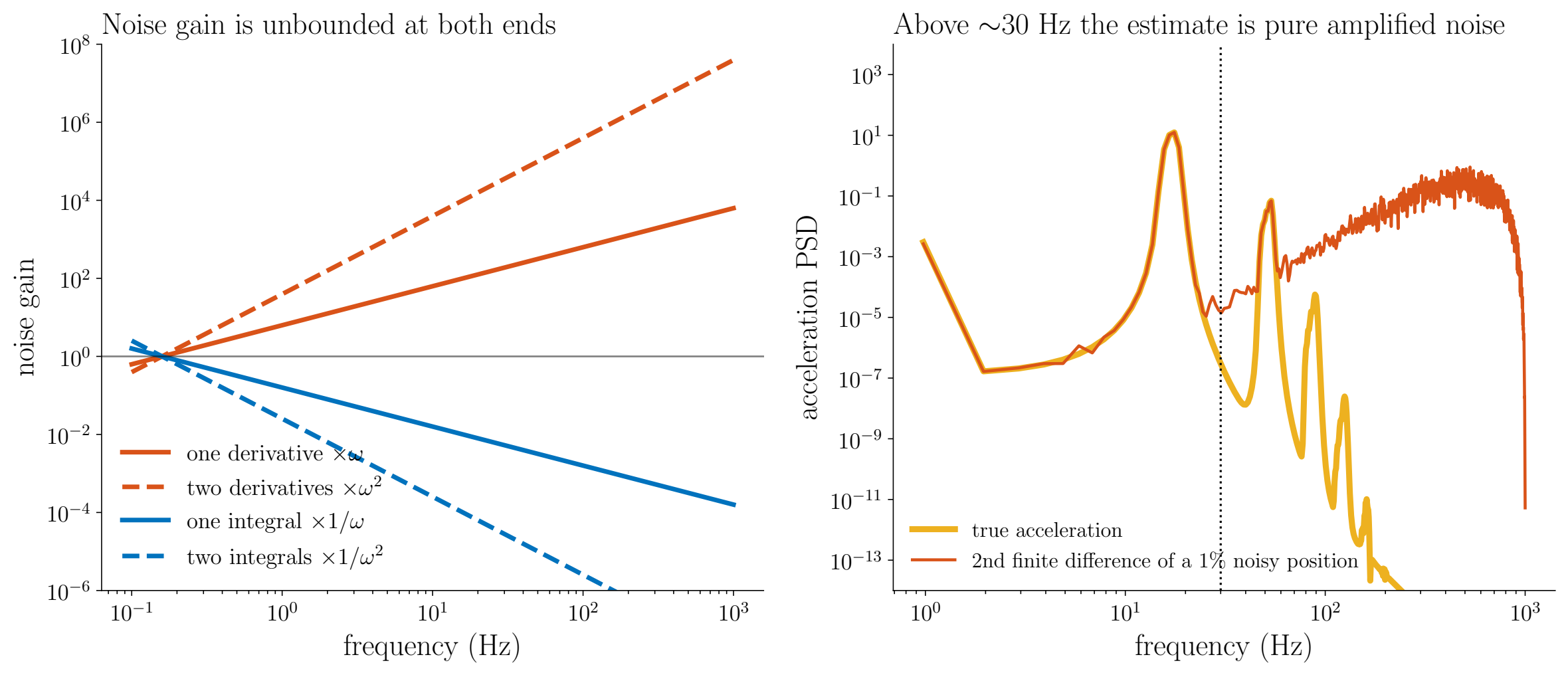

Noise gain is unbounded at both ends
noise gain
frequency (Hz)
one derivative ×ω
two derivatives ×ω²
one integral ×1/ω
two integrals ×1/ω²
Above ∼30 Hz the estimate is pure amplified noise
acceleration PSD
frequency (Hz)
true acceleration
2nd finite difference of a 1% noisy position